

Smart as Shit: Finding Data in Unexpected Places

What wastewater, broken traffic counters, and systems thinking can teach us about measuring the real world

Mason Perry • NULLWORKS • Working paper, September 2026

Abstract

Tourism-dependent communities often need population estimates at timescales that census counts cannot provide. This case study reconstructs an applied measurement approach developed in Big Bear, California, where conventional visitor-counting methods were unreliable. Wastewater influent was treated as an unavoidable physical trace of human presence, then tested against weather to determine when flow represented people and when it represented hydrology. Monthly Big Bear Area Regional Wastewater Agency (BBARWA) influent data from January 2018 through August 2026 were merged with NOAA GHCN-Daily observations for Big Bear Lake. July influent remained tightly bounded at 1.73–1.96 million gallons per day (MGD) across nine summers, with a mean of 1.851 MGD and coefficient of variation of 4.7%, making July a comparatively stable sanitary/tourism control. March behaved differently: March influent ranged from 1.87 MGD in dry 2018 to 5.13 MGD in 2023, and October–March precipitation correlated strongly with March influent (Pearson $r = 0.94$, $n = 9$). The relationship remained strong when either 2019 or 2023 was removed and remained positive when both wettest winters were removed. Member-agency shares stayed broadly stable during spring spikes, supporting a system-wide infiltration and inflow (I&I) interpretation rather than a single tourism-corridor effect. A descriptive shoulder-season analysis suggests somewhat higher fall loading after 2020, but the effect is sensitive to weather screening and the short pre-period prevents causal attribution to tourism marketing. The central result is methodological: wastewater can function as a population proxy only when the mechanisms that also move wastewater are explicitly modeled. In spring the sewer was not counting tourists. It was counting the thaw. In July it was counting people.

Keywords: wastewater flow; transient population; tourism measurement; infiltration and inflow; snowmelt; proxy data; Big Bear; systems thinking

1. Introduction

Measurement problems are often framed as instrumentation problems: if a phenomenon is difficult to count, install a better counter. That instinct fails when the available counters are unreliable, incomplete, expensive, or measure the wrong thing. In tourism-dependent communities, visitor volume is a particularly difficult quantity because the relevant population is transient. Hotel occupancy excludes day-trippers and guests staying in private homes. Card-spend data count transactions rather than bodies and are shaped by payment behavior. Traffic counters can fail mechanically or algorithmically and still produce precise-looking numbers.

The original Big Bear consulting problem was practical rather than academic: how many people were actually coming to the mountain, when were they coming, and was the tourism strategy filling historically weak shoulder periods? After conventional traffic-counting outputs proved unreliable, the problem was reframed around an unavoidable behavior. Visitors may skip a hotel, a restaurant, a ski resort, or a retail transaction. They do not indefinitely skip sanitation. The regional wastewater system therefore offered a long-running, operationally maintained signal of human presence.

The idea is consistent with a broader literature on wastewater-based population estimation. Daughton proposed sewage-borne biomarkers as a real-time basis for estimating small-area populations [1]. Later work has used wastewater flow, chemical parameters, ammonia, and other markers to estimate de facto or dynamic populations [2–5]. However, the same literature emphasizes a critical limitation: wastewater is not composed only of sanitary discharge. Rainfall, groundwater, and snowmelt can enter collection systems through infiltration and inflow (I&I), changing plant flow without any change in people [6,7]. In a mountain community with snow accumulation and spring thaw, this is not a minor correction. It is the central confounder.

This study asks a narrow question first: are the large spring wastewater peaks in Big Bear plausibly visitor-driven, or are they weather-driven I&I? It then uses the answer to define when wastewater is a credible population proxy and when it is not. A secondary analysis asks whether fall shoulder-season loading appears higher after 2020, while deliberately avoiding causal claims that the available data cannot support.

2. Case context

BBARWA provides centralized wastewater conveyance, treatment, and disposal for the Big Bear Valley. Its member systems include the City of Big Bear Lake, Big Bear City Community Services District, and San Bernardino County Service Area 53B; the agency reports that the regional treatment system serves the full Big Bear Valley and that each member agency operates its own collection system before delivering flow to BBARWA [8]. The plant treats domestic and commercial wastewater and reports substantial seasonal variation in flow [9].

That system architecture matters for this study. A regional plant integrates human activity across multiple communities, while the member shares provide a coarse spatial check on whether an anomalous spike is concentrated in one visitor-heavy service area or distributed across the network. The wastewater plant is therefore both a sensor and a mixing chamber: useful precisely because it aggregates the valley, dangerous if every gallon is naively treated as a person.

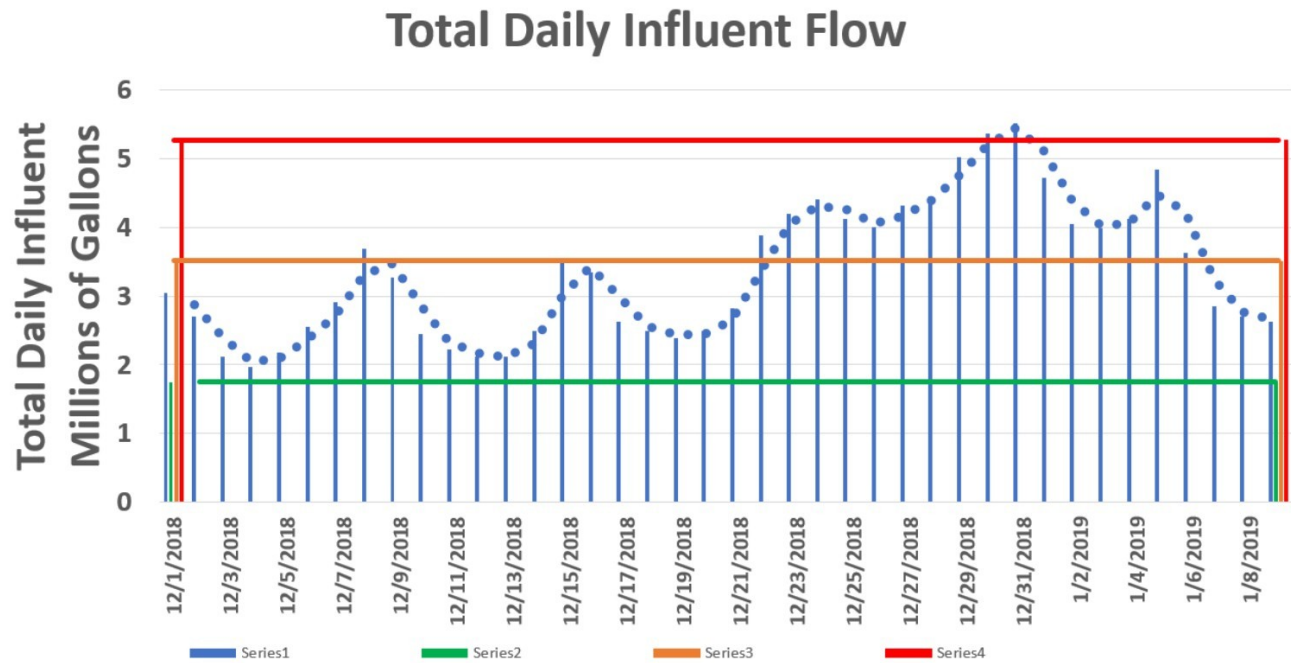


Figure 1. Archived 2018–2019 working graph comparing observed daily influent with independently estimated population-load bands derived from permanent-resident and rental-capacity assumptions. The underlying calculation worksheet is no longer available, so this figure is treated as historical context only and is excluded from inferential statistics.

3. Data

3.1 Wastewater flow

The reconstructed analytical database contains BBARWA influent observations spanning January 2018 through August 2026. The study uses monthly average influent in million gallons per day (MGD), with daily data retained where available in the underlying workbooks. For March, member-agency contribution shares are available for the City of Big Bear Lake, Big Bear City CSD, and Fawnskin/CSA 53B.

3.2 Weather

Daily weather observations were taken from NOAA's Global Historical Climatology Network Daily (GHCN-D) station USC00040741, BIG BEAR LAKE, CA US, at approximately 34.243° N, 116.917° W and 6,753 ft elevation [10]. Variables used were precipitation, snowfall, snow depth, daily maximum temperature, and daily minimum temperature. The weather file contains 3,135 study-window days; precipitation is missing on 41 days, snowfall on 89, snow depth on 128, and TMAX on 23. Monthly totals use available observations. The official daily file used here extends through August 9, 2026.

4. Methods

The analysis treats total plant influent as a composite signal:

$$\text{Observed influent} = \text{resident sanitary load} + \text{transient sanitary load} + \text{I\&I} + \text{noise}$$

The goal is not to estimate each term perfectly. It is to identify periods in which one term clearly dominates enough to make interpretation safer.

4.1 July control

July was selected as the primary people-dominated comparison month because it is a peak visitor period but largely outside the spring snowmelt regime. For each year, July average influent was calculated and summarized by range, mean, standard deviation, coefficient of variation, and linear year trend. This is not an assertion that July flow is a census. It is a same-system control expected to contain sanitary loading with relatively less spring hydrologic contamination.

4.2 March weather test

For each March from 2018 through 2026, average March influent was compared with: March precipitation; March snowfall; October–March cumulative precipitation; October–March cumulative snowfall; February+March precipitation; average March maximum temperature; and thaw degree-days, defined as the monthly sum of $\max(\text{TMAX} - 32^\circ\text{F}, 0)$. Pearson correlation was used as a descriptive measure of association. With only nine March seasons, correlations are interpreted as pattern strength rather than definitive causal estimates.

4.3 Lag and robustness checks

To test a mechanism consistent with infiltration and snowmelt, same-month precipitation was compared with a two-month precipitation measure (current month plus prior month). The primary March correlation was then recalculated after excluding 2019, after excluding 2023, and after excluding both years. This deliberately attacks the possibility that two extreme wet winters are manufacturing the entire relationship.

4.4 Spatial/member-share check

March member shares were inspected across years. A tourist-corridor explanation would be more plausible if extreme March flow years also showed a major shift toward one member system. A stable distribution during spike years is more consistent with a network-wide hydraulic effect.

4.5 Exploratory three-layer decomposition

A conservative lower-bound decomposition was constructed for visualization. Each year's dry-weather floor was defined as the lowest monthly flow from June through October. During February–May, sanitary/seasonal flow was capped at that same year's July value; flow above July was labeled lower-bound I&I. Outside February–May, no I&I amount was assigned by this model. The decomposition is intentionally asymmetric: it can understate I&I, but it is designed not to manufacture spring I&I from small seasonal differences.

4.6 Shoulder-season test

Fall flow was evaluated separately from the spring analysis. September–November and October–November averages were calculated for each year and normalized to the same year's July value. 2018–2019 were used as the available pre-2020 baseline; 2021–2025 were used as the later comparison period; 2020 was displayed but excluded from the before/after average. Because fall

rainfall can also affect sewer flow, a sensitivity analysis filtered shoulder months by two-month precipitation thresholds from 1.0 to 3.0 inches. The analysis is descriptive and cannot identify the effect of tourism advertising.

5. Results

5.1 July behaves like a stable sanitary/tourism control

Across 2018–2026, July influent remained between 1.7308 and 1.9591 MGD. The nine-year mean was 1.851 MGD, with a standard deviation of 0.088 MGD and a coefficient of variation of approximately 4.7%. The linear slope was approximately -0.0058 MGD per year, and the correlation between year and July flow was weak ($r \approx -0.18$). The practical result is more important than the sign of the trend: peak-tourism July is almost a flat line.

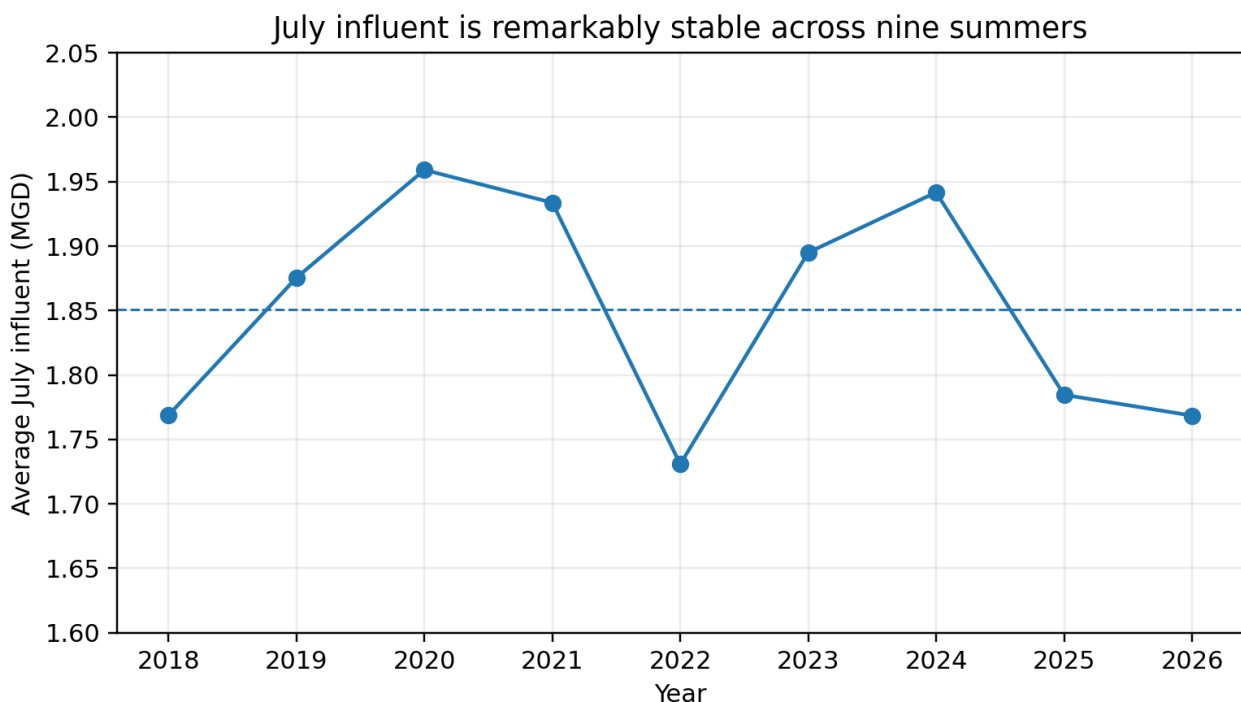


Figure 2. Average July influent by year. July remains within a narrow band across nine summers.

5.2 March is a weather year

March flow was not stable. It ranged from 1.8711 MGD in 2018 to 5.1294 MGD in 2023. The two largest March sewer years, 2019 (4.4804 MGD) and 2023 (5.1294 MGD), were also the two wettest October–March seasons in the nine-year window at 33.23 and 31.34 inches of precipitation, respectively. By contrast, 2018 accumulated only 8.02 inches of October–March precipitation and March flow was 1.8711 MGD—essentially the July sanitary range.

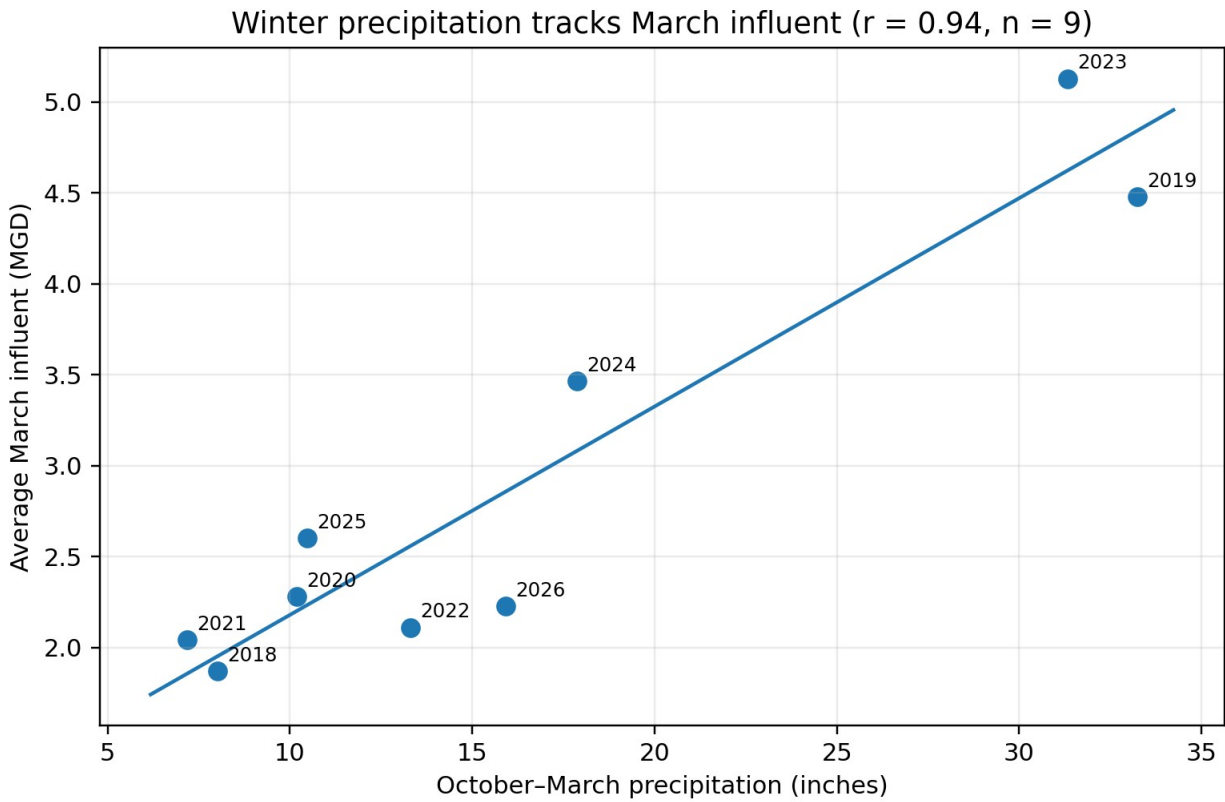


Figure 3. October–March precipitation versus March influent. Pearson $r = 0.94$ ($n = 9$).

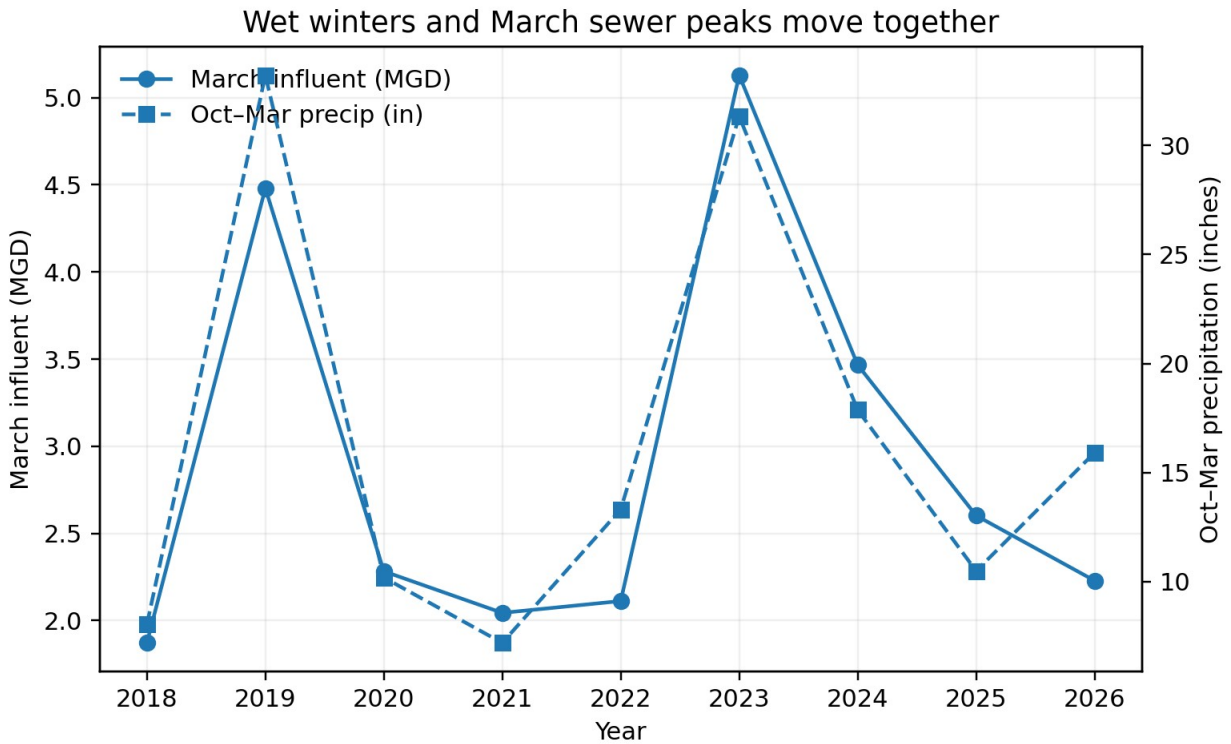


Figure 4. Annual comparison of March influent and preceding October–March precipitation.

Table 1. March flow and winter weather by year

Year	March MGD	Oct–Mar precip (in)	Oct–Mar snow (in)
2018	1.87	8.02	17.7
2019	4.48	33.23	62.3
2020	2.28	10.18	59.5
2021	2.04	7.18	49.8
2022	2.11	13.31	12.5
2023	5.13	31.34	83.4
2024	3.47	17.87	34.2
2025	2.60	10.48	29.8
2026	2.23	15.91	25.4

5.3 The relationship survives obvious stress tests

The primary October–March precipitation versus March-flow relationship was $r = 0.939$ across all nine seasons. Excluding 2019 yielded $r = 0.935$; excluding 2023 yielded $r = 0.918$. Excluding both wettest winters reduced the range of the remaining data substantially, but the relationship remained positive at $r = 0.696$ ($n = 7$). The result is therefore not a one-outlier trick.

Other comparisons support a delayed hydrologic mechanism. February+March precipitation versus March flow was $r \approx 0.86$, and October–March snowfall versus March flow was $r \approx 0.75$. Across all months, same-month precipitation correlated with flow at about $r = 0.62$, while current-plus-prior-month precipitation correlated at about $r = 0.82$. By contrast, July–August precipitation versus July–August flow was essentially uncorrelated ($r \approx -0.07$). The plant appears to respond strongly to antecedent cold-season water and much less to summer rain.

5.4 Temperature alone is not the switch

March thaw degree-days were negatively correlated with March flow ($r \approx -0.53$). At first glance this seems contradictory, but it is physically plausible: the wettest winters can remain colder longer. Heat without stored water has little to release. A warm, dry March need not create large I&I, while a colder, water-loaded season can sustain elevated groundwater and snowmelt inputs. The useful state variable is therefore not temperature alone but available water interacting with thaw and collection-system leakage.

5.5 The spring spike is system-wide

Member shares did not flip toward a single service area in high-flow Marches. Across 2018–2026, the City of Big Bear Lake contributed roughly 53–62% of March flow, Big Bear City CSD roughly 35–44%, and Fawnskin/CSA 53B roughly 3–5%. The broad stability of these shares during 2019 and 2023 supports a network-wide hydraulic explanation rather than a single hotel or resort corridor.

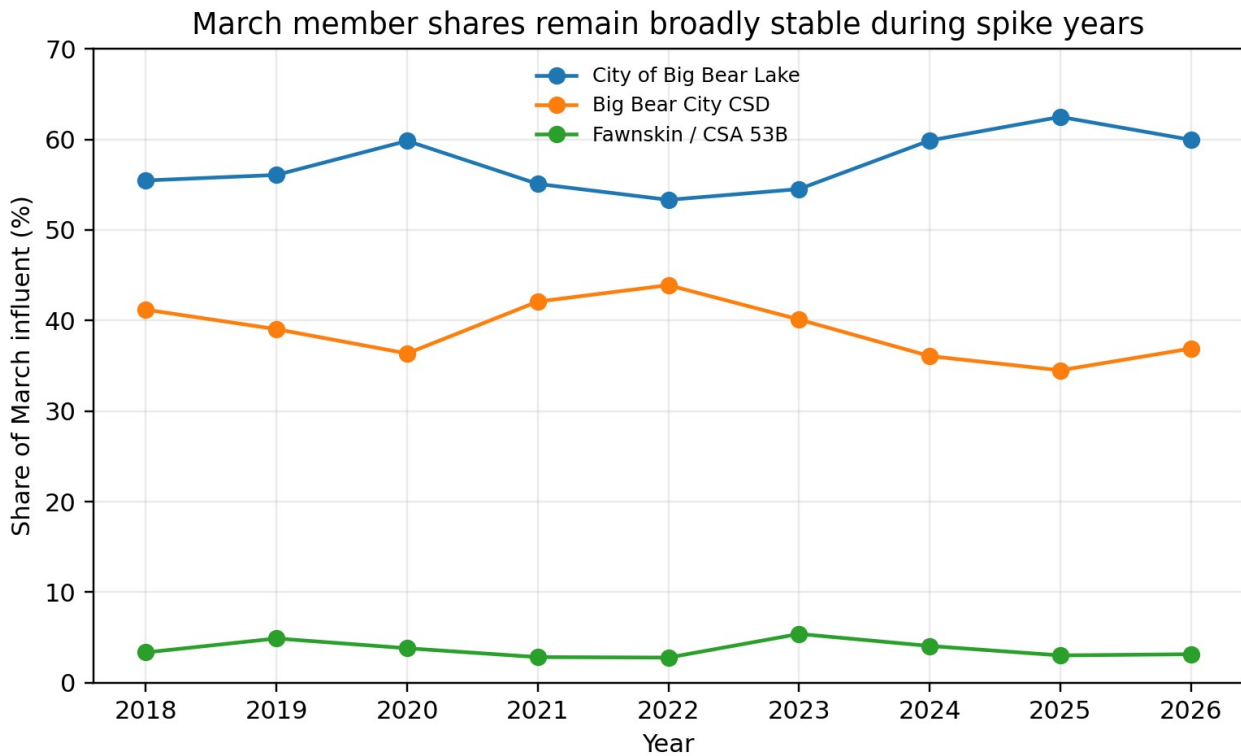


Figure 5. March member-agency shares by year. The distribution remains broadly stable even when total March flow spikes.

5.6 The shoulder-season signal is suggestive, not proved

Raw fall averages show some evidence of higher later-period loading. Average September–November flow for 2021–2025 was approximately 5.4% above the 2018–2019 average, while October–November was approximately 9.1% higher. When normalized to each year's July control, the raw September–November post/pre uplift was approximately 3.5%.

The result is not robust enough for a causal tourism-marketing claim. Weather screening changes the estimate materially and leaves small samples: under two-month precipitation caps from 1.0 to 3.0 inches, the July-normalized post/pre uplift ranges from about 2% to 7%. At the 2.0–3.0 inch screens, the uplift is only about 2.1–2.3%. The year pattern is also not monotonic: 2024 dips before 2025 rebounds. The available 2018–2026 data therefore support a shoulder-season hypothesis, not a conclusion.

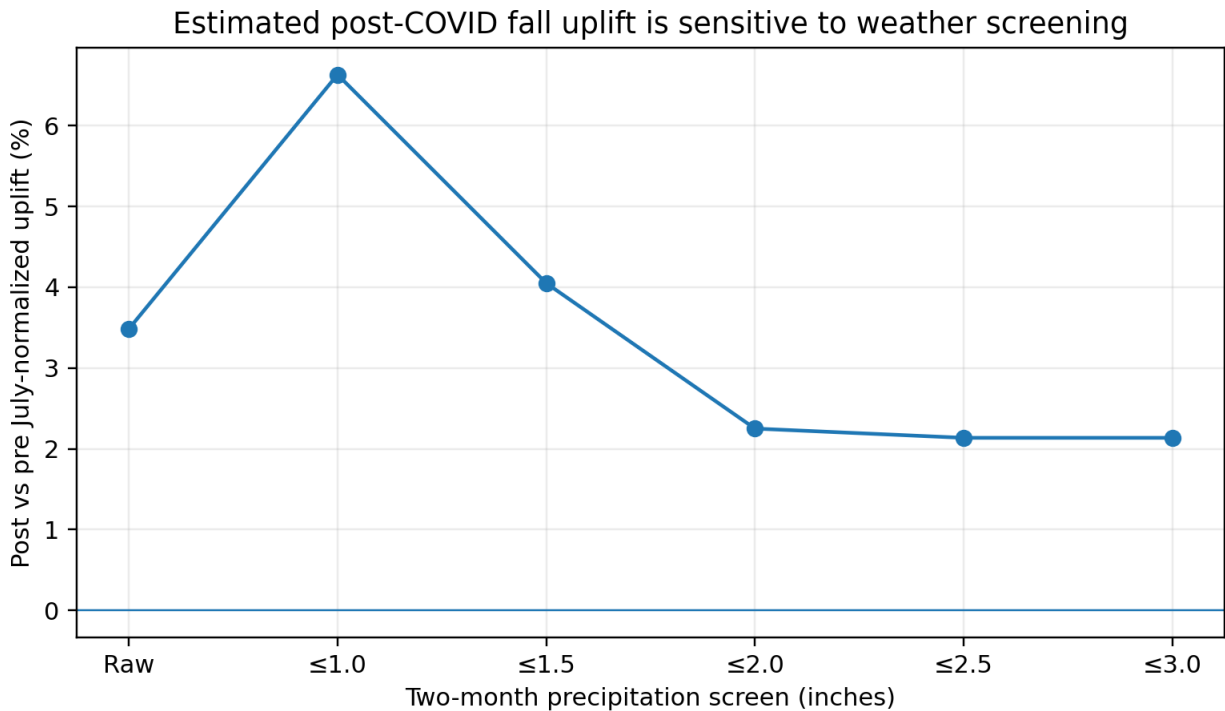


Figure 6. Sensitivity of the July-normalized fall uplift to precipitation screening. Small sample sizes make these estimates descriptive only.

6. Discussion

6.1 A sewer can be two different sensors

The central mistake in using wastewater flow as a population proxy is to assume that the proxy has one meaning. It does not. During dry or frozen-ground periods, changes in flow can be dominated by sanitary load and therefore contain useful information about human presence. During wet or thaw periods, extraneous water can dominate. EPA guidance explicitly separates sanitary base flow, groundwater infiltration, and rainfall-derived inflow/infiltration [6], and cold-climate sewer research documents both direct snowmelt inflow and snowmelt-induced groundwater infiltration [7]. The Big Bear data reproduce that mechanism at an unusually intuitive scale.

This explains why the same instrument can be useful in July and misleading in March. The sensor is not broken; the measurement model is incomplete. The correct question is not 'Does wastewater measure population?' but 'Under what operating conditions is population the dominant component of wastewater flow?'

6.2 The control is July

The strongest design feature of this case is not the March correlation by itself. It is the contrast with July. If the multi-MGD March peaks were primarily caused by more people, a comparable structural expansion should be visible in peak-tourism July. It is not. Nine Julys remain packed into a 0.23 MGD range. Meanwhile March more than doubles between dry and wet years. That

cross-season control turns a visually interesting correlation into a much stronger systems argument.

6.3 Lag is mechanism-consistent

The stronger two-month precipitation relationship relative to same-month precipitation is also important. Direct visitor wastewater should respond almost immediately to occupancy. Groundwater infiltration and snowmelt do not have to. Water can accumulate as snow, raise groundwater tables, migrate through soils, enter cracked pipes or joints, and continue loading the collection network after a storm or melt event. The observed lag therefore fits the physical mechanism expected for I&I. It does not prove a specific hydrograph without daily plant data and snow-water equivalent, but it points in the correct direction.

6.4 Why the old population-load graph still matters

The archived 2018–2019 holiday graph cannot be used as a formal validation test because the underlying resident/rental assumptions are no longer available. It is still useful as provenance. It documents that the wastewater-as-population idea was not invented after seeing the 2018–2026 weather correlation. The original operational analysis already treated holiday wastewater loading as a way to bracket transient population. The new analysis improves that old method by showing when the same signal must be rejected as hydrology.

6.5 The broader systems lesson

The case illustrates a general approach to real-world measurement: when the obvious sensor fails, look for the unavoidable physical trace produced by the system. Soil organisms decompose cotton. People generate wastewater. Vehicles leave tire counts. Warehouses draw power. Networks emit logs. The proxy does not need to be glamorous; it needs to be unavoidable, recorded, and interpretable.

But proxy discovery is only half the job. Every proxy has alternate causes. The analyst must ask what else can move the signal, identify a control condition, test the suspected confounder, and preserve uncertainty. A good proxy is not one that correlates once. It is one whose failure modes are understood.

7. Limitations

This study has several limitations. First, the March analysis contains only nine seasons. The correlation is strong but the sample is small, and the observations are not a randomized experiment. Second, weather is represented by one NOAA station; spatial variation in snowpack and precipitation across the 79,000-acre valley is not captured. Third, the weather record has missing observations, and the 2026 daily file ends on August 9. Fourth, monthly flow obscures the precise lag structure that daily plant influent would reveal. Fifth, snowfall is not equivalent to snow-water equivalent; direct SWE observations would improve the hydrologic model. Sixth, the archived population-load graph lacks the original resident/rental calculation and therefore cannot support an absolute population-error estimate. Seventh, the shoulder-season before/after comparison has only two pre-2020 years and is sensitive to weather filtering. It cannot identify the effect of advertising, lodging supply, remote work, pandemic-era behavior, or other concurrent changes.

The conservative decomposition is exploratory rather than a calibrated hydrologic model. It intentionally labels only spring flow above the same year's July control as lower-bound I&I and therefore almost certainly leaves some I&I inside the sanitary/seasonal layer. That bias is preferable for the present purpose because the goal is to avoid over-attributing flow to weather.

8. Conclusion

The Big Bear case began with a mundane measurement failure: the community wanted to know how many people were showing up, and the obvious counters were not dependable. Wastewater offered a better trace because nearly everyone contributes to it whether or not they stay in a hotel, swipe a card, or pass through a functioning traffic counter.

The reconstructed 2018–2026 data show why that insight works only with guardrails. July influent is remarkably stable across nine peak-tourism summers. March is not. Winter precipitation explains the spring excursions extremely well, the relationship survives removal of either extreme wet year, prior-month water improves the relationship, and member-agency shares remain broadly stable through the spikes. Those findings are consistent with region-wide infiltration and inflow driven by winter water and thaw.

The practical rule is simple: treat wastewater as a context-dependent sensor. In dry or frozen-ground periods, it can be a useful proxy for transient human presence. In spring, it may be telling you more about the mountain than the tourists.

In spring the sewer was not counting tourists. It was counting the thaw. In July it was counting people.

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Appendix A. March test data

The table below reproduces the core March comparison used in the analysis. Values are monthly aggregates from the merged BBARWA/NOAA research workbook.

Year	March MGD	Mar precip	Mar snow	Oct–Mar precip	Oct–Mar snow	Thaw DD
2018	1.87	4.34	4.1	8.02	17.7	555
2019	4.48	3.21	7.0	33.23	62.3	454
2020	2.28	4.32	5.0	10.18	59.5	393
2021	2.04	1.43	5.3	7.18	49.8	522
2022	2.11	1.99	4.2	13.31	12.5	749
2023	5.13	11.88	39.5	31.34	83.4	332
2024	3.47	4.04	9.0	17.87	34.2	445
2025	2.60	4.57	16.3	10.48	29.8	577
2026	2.23	0.00	0.0	15.91	25.4	1041

Core correlations from the working analysis: Oct–Mar precipitation vs March MGD $r = 0.94$; Feb+Mar precipitation vs March MGD $r = 0.86$; Oct–Mar snowfall vs March MGD $r = 0.75$; March thaw degree-days vs March MGD $r = -0.53$. Robustness: $r = 0.93$ excluding 2019; $r = 0.92$ excluding 2023; $r = 0.70$ excluding both.

Appendix B. Data and analysis provenance

Primary flow analysis workbook: BBARWA_Smart_as_Shut_Research_Workbook_v2.xlsx. The workbook contains the merged monthly series, March test, source notes, three-layer decomposition, fall shoulder test, robustness checks, weather-adjusted shoulder sensitivity, and the claims/figure map used to draft this paper. The historical Figure 1 image was provided separately from the author's prior consulting work.